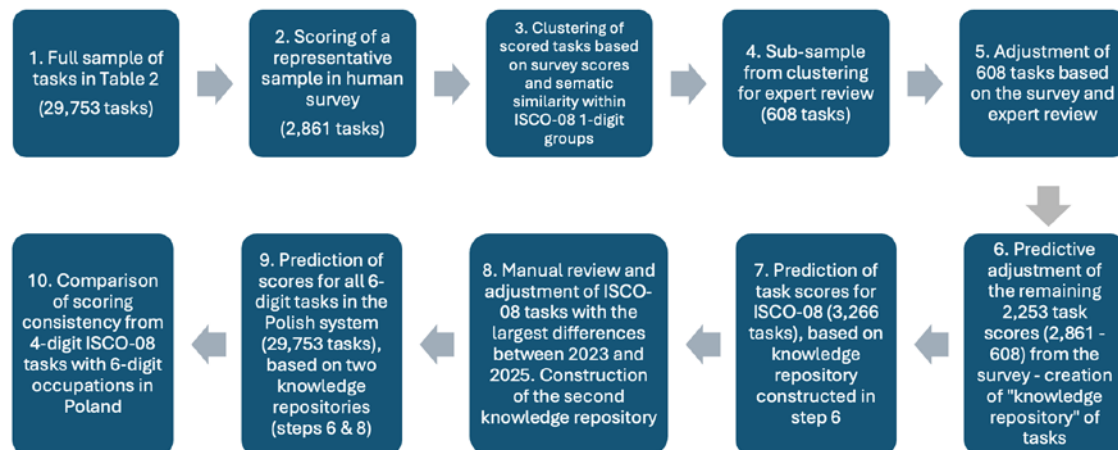


► 4 Expert validation survey

The expert group includes international specialists at the ILO and its partner organizations, NASK and the Polish Ministry of Family, Labour and Social Policy.⁹ The primary criterion for expert selection is their understanding of labour markets, including the complexity of occupations and tasks, and their real-life applications. While the experts are expected to have familiarity with the current family of generative AI technologies and their applications, expertise in AI design or development is not a requirement. This approach ensures that the focus remains on the feasibility and practical implications of automation within existing technological constraints, as technology experts tend to overstate technological capacities and risks when evaluating broader applications, particularly in workplace settings (Karger et al., 2023). This tendency is evident in earlier automation studies, which often assigned high displacement potential to routine tasks or entire occupations, such as garment production. As de Mattos et al. (2020) illustrate, these technological predictions frequently failed to account for the challenges of automating tasks requiring high dexterity or handling complex materials, such as stitching pliable fabrics. Accordingly, the assessment of automation potential in our study prioritizes expertise of task execution and workplace realities over purely technological expertise.

Since the experts cannot review all 2,861 tasks used in the survey, we focus on a representative sub-sample, selected in such a way to subsequently permit adjustments to entire task clusters. The overall scheme of this score adjustment procedure is presented in Figure 11.

► Figure 11. Schema of the scoring stages



First, we cluster the tasks into four fixed intervals, based on the distribution of the mean scores from the survey.¹⁰ We then use hierarchical clustering to group similar tasks within those clusters based on their semantic content. First, we assign embeddings to each task description using the *text-embedding-ada-002* model from OpenAI. Next, for each subgroup – defined by the combination of an occupational group at ISCO-08 1-digit level and a particular survey score cluster – we

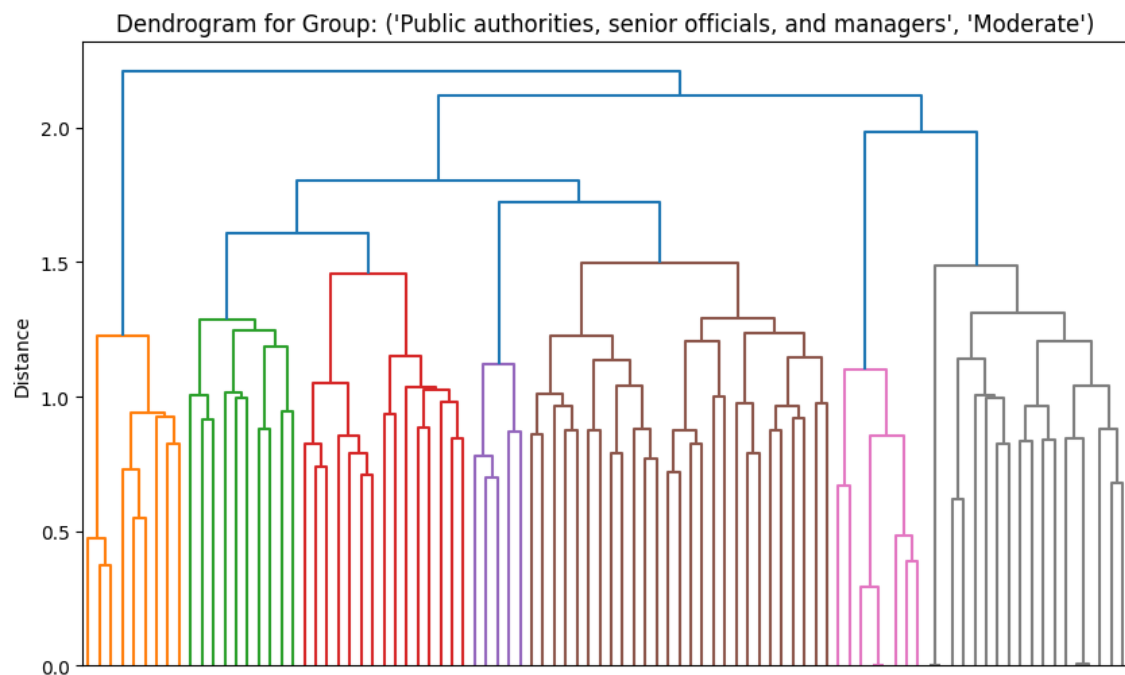
⁹ A full list of contributors is provided in the acknowledgements section.

¹⁰ $\mu \in [0, 0.1] \cup (0.1, 0.3] \cup (0.3, 0.5] \cup (0.5, 1]$

perform Ward’s hierarchical clustering.¹¹ We generate a dendrogram for each subgroup, plotting the tasks along the horizontal axis and the linkage distances (or “heights”) along the vertical axis. Clusters are then formed by “cutting” the dendrogram at a chosen linkage height, such that every branch below this cut line becomes its own cluster (Figure 12).

In our approach, we select the height threshold manually for each subgroup through visual inspection of the dendrogram. This allows us to flexibly accommodate heterogeneity among subgroup’s tasks, so as to ensure that each final cluster contains tasks that are genuinely similar in content. Once the distance threshold is determined, all tasks in the subgroup are assigned to a unique cluster. The process is repeated for each occupation–score–cluster combination. Although our visual inspection introduces an element of subjective judgment, it ensures that spurious or extremely fine-grained splits are avoided, helping to maintain coherent groupings of tasks for the subsequent expert review.

► **Figure 12. Example of a dendrogram used for the review of tasks’ semantic clustering**



Once each task had been assigned to a cluster, based on its 1-digit occupational group, scoring bracket from the survey and the semantic similarity of tasks, we generate the final sample of tasks for expert review. To ensure coverage across all clusters while limiting the total burden on experts, we applied a scaling factor (set to 20%) to determine the approximate fraction of tasks to sample from each group.¹² We then enforced a minimum of one task per cluster (to avoid excluding smaller clusters entirely) and a maximum of five tasks per cluster to keep the workload manageable. Through this procedure, we achieve a balanced, yet compact, sample that captures semantic diversity while preventing over-sampling from any single cluster. We retain the cluster assignment for each task, so that the expert review can be subsequently looped back into adjustments of the scores in the cluster that the sampled task originated from.

¹¹ Ward’s method seeks to minimize the variance within each cluster, making it particularly suitable for tasks with varied textual content.

¹² Specifically, the formula took the size of each cluster relative to the group total, multiplied by the scaling factor.

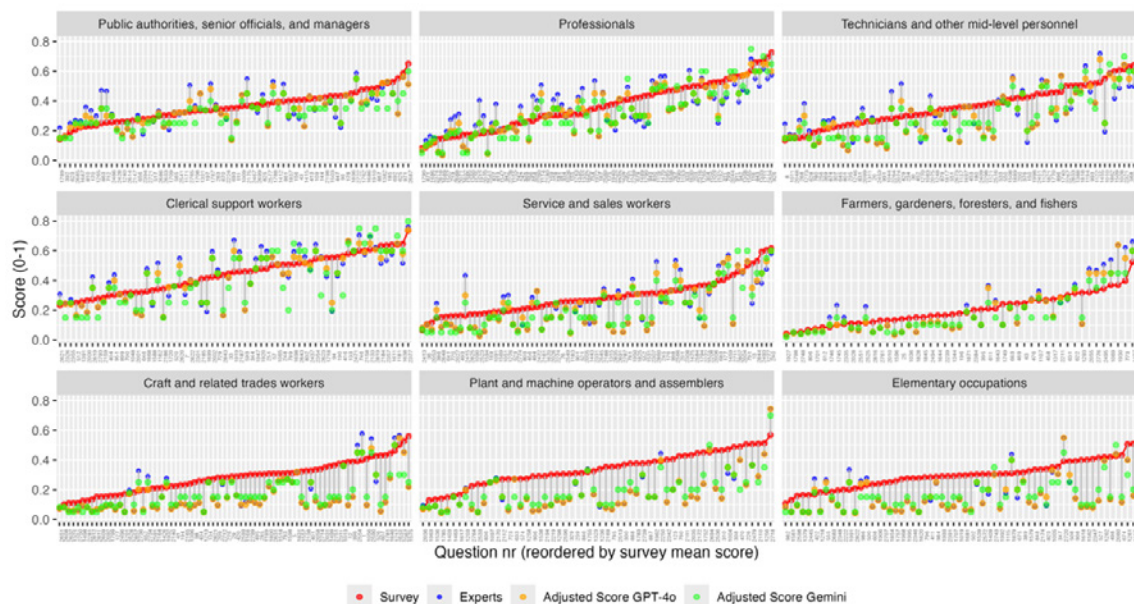
Each of the selected 608 tasks was then presented to a panel of experts. For each task, the experts were shown (i) the occupation and cluster information, (ii) the average survey score (rescaled to 0–1), and (iii) the full task description. If they agreed with the survey score, they left it unchanged; otherwise, they entered a revised expert score (0–1).

In many cases, the differences between the survey and expert scores are non-trivial. They also vary significantly across occupational groups, revealing distinct patterns based on the nature of pertaining tasks (Figure 13). In occupations dominated by manual and physical tasks, such as Craft and Related Trades Workers, Plant and Machine Operators, and Farmers, Gardeners, Foresters, and Fishers, experts consistently assigned lower scores compared to the survey respondents, reflecting the practical limitations of automating tasks that require physical manipulation or manual dexterity. In contrast, occupations involving more cognitive or knowledge-intensive tasks, such as Clerical Support Workers and Professionals, show a much more mixed picture. While some tasks in these groups received lower scores, many others were revised upwards by the expert group, reflecting the greater adaptability of generative AI for structured, data-driven, or communication-focused tasks. For example, Clerical Support Workers display a combination of increases and decreases, suggesting nuanced task-specific assessments where automation potential depends heavily on the nature of individual tasks.

► 5 Adjustment of survey sub-sample

In the next step, we compared the average of expert scores with the original mean survey scores using two independent AI models as impartial arbiters to propose an adjusted score.¹³ The models were informed about the sources of the scores in such a way that would prevent them from automatically siding with the expert group: group one was defined as *“practical knowledge of the work and types of tasks they were presented, since they worked in the same occupational groups from which we selected tasks they scored”*, while the second group was described as *“respondents with more theoretical knowledge of technology and labour markets”*. Rather than simply averaging the two scores, the models were explicitly instructed to focus their analysis on the task characteristics and the current capabilities of generative AI and required to provide justifications for each adjustment and its relative position to the original scores.

► Figure 13. Task-level scores from the survey and experts, compared to AI-arbitrated scores



The overall correlation between the adjusted scores produced by the two AI models is 0.96, with correlations within occupational groups exceeding 0.9 in all but one case¹⁴ (Figure 14). This high level of agreement demonstrates that, when provided with sufficient background knowledge in the form of reference ranges of scores, the models converge on remarkably similar predictions. Notably, only three tasks exhibited an absolute difference in scores exceeding 0.2 (orange dots in Figure 13). In these cases, GPT-4o tended to adopt a more optimistic perspective on the feasibility of automation, whereas Gemini emphasized the importance of human judgment as a critical factor.

¹³ We used OpenAI's GPT-4o and Google's Gemini Flash 1.5, which have comparable abilities. We also tested OpenAI o1, which at the time of writing was among the models with strongest capacities of « algorithmic reasoning » and observed very similar results. The process was set up as a cycle of sequential API calls, which loop over each task and capture the response into a data frame.

¹⁴ Public authorities, senior officials and managers in Figure 10.